TYPED MEASUREMENT-TRIGGERED TEMPORAL MEMORY FOR TECHNICAL MONITORING

Technical field

[0001] Technical monitoring of asynchronous multichannel sensor streams.

Background

[0002] Padding and resampling can process channels that have no new measurement and obscure channel identity, measurement absence and elapsed time.

Summary

[0003] A monitor updates addressed channel statistics only on actual measurements, applies channel-typed comparisons before mixing, and combines the result with timestamped temporal memory.

Brief description of the drawings

[0004] FIG. 1 is a block diagram of an implementation. FIG. 2 shows operations of the implementation. Reference signs identify the same functional elements throughout the drawings.

Detailed description

[0005] In FIG. 1, 100 identifies measurements; 110 identifies channel slots; 120 identifies comparators; 130 identifies temporal state; 140 identifies monitor; 150 identifies training. In FIG. 2, 210 identifies measurement, 220 identifies slot write and 230 identifies timed query.

[0006] Input 100 supplies records (t,c,x) identifying a sensor channel c. State store 110 contains a count, mean, second-moment accumulator, last value and last measurement time

per channel. Unmeasured channels do not acquire fabricated observations. An access can compute staleness from the current causal cutoff and stored timestamp.

[0007] For a new value, an example update is $n' = n+1$, $\delta = x-\mu$, $\mu' = \mu+\delta/n'$ and $M_2' = M_2 + \delta(x-\mu')$. The updated sufficient statistics and latest value remain in the addressed channel slot. Alternative bounded-window or exponentially weighted statistics can be used with their own timestamps.

[0008] Comparator 120 applies learned thresholds or nonlinear comparisons specific to channel type before cross-channel mixing. A threshold feature may be $\sigma(\alpha_c(x-b_c))$. Distinct channels do not silently share a numerical threshold when their units differ. Missingness and staleness are supplied as explicit inputs.

[0009] Temporal engine 130 transports memory modes over the actual measurement gap and combines the updated channel features with persistent state. Only the addressed channel statistics are written on the measurement. A separate scheduled query may read the monitor during silence without pretending that a measurement arrived.

[0010] Output 140 provides a device-state estimate, machine alarm or monitoring signal. A shared temporal core may serve multiple installations through domain-specific channel maps, scaling and output interfaces. Training uses observed data and declared labels; a changed channel map is stored with the deployed model.

[0011] FIG. 1 shows the event input 100, channel store 110, typed comparison 120, temporal engine 130, monitoring output 140 and fitting processor 150. FIG. 2 separates measurement arrival 210, addressed update 220 and scheduled monitoring query 230. A hardware embodiment banks channel slots in SRAM and enables writes only for a received channel; missing-data read traffic is counted separately.

Hardware implementation

[0012] A digital implementation may use synthesizable register-transfer logic mapped to an

FPGA or to an application-specific integrated circuit. Addressed state uses register banks or SRAM macros; arithmetic uses integer, fixed-point or floating-point datapaths. Ready/valid or request/acknowledge interfaces apply backpressure without accepting a message twice. Synchronous FPGA execution, locally clocked circuits and asynchronous circuits are alternative implementations, each preserving its declared causal ordering and numerical precision.

[0013] Before deployment, a reference trace can check selected addresses, values, timestamps, publication versions and output probabilities against the circuit at the declared numerical precision. Timing quantization, ties, cancellation, overflow and saturation are specified for the implementation. FPGA resource utilization, post-layout timing and measured energy are implementation-dependent quantities; no particular measured hardware result is required by this embodiment.

CLAIMS

1. A method of operating a technical sensor monitor, comprising receiving asynchronous measurements identified by channel and time; maintaining addressed per-channel sufficient statistics and last-measurement timestamps; writing a channel slot in response to a measurement on that channel; applying channel-typed threshold comparisons before cross-channel mixing; integrating resulting features in temporal memory using actual elapsed time; and issuing a monitoring signal.

2. The method of claim 1, wherein the sufficient statistics comprise count, running mean and a second-moment accumulator.

3. The method of claim 1, wherein staleness and measurement absence are represented separately from measured values.

4. The method of claim 1, wherein a scheduled query reads temporal state without writing a fabricated sensor measurement.

5. The method of claim 1, wherein a shared temporal core receives domain-specific channel mappings and scaling parameters.

6. A sensor-monitoring apparatus comprising a measurement input, addressed statistic storage, a typed comparator and a temporal processor configured to perform the method of claim 1.

7. A non-transitory computer-readable storage medium storing instructions that cause a processor to perform the method of claim 1.

ABSTRACT

A technical monitor receives asynchronous measurements with channel identifiers and timestamps. Addressed channel slots store sufficient statistics and last-measurement times and are written only when measurements arrive on their channels. Typed comparisons precede cross-channel mixing. A temporal-memory engine integrates the resulting features using elapsed time and explicit missingness information. Scheduled queries can produce monitoring outputs during silence without manufacturing new measurements. A shared core may use installation-specific channel maps.